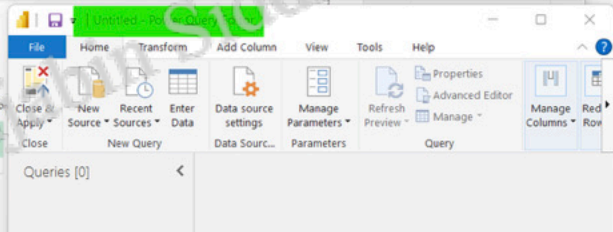
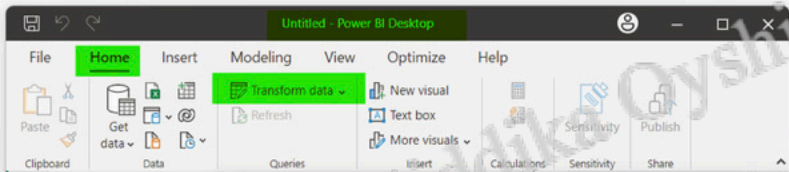


Launching Power Query Editor

To open the Power Query Editor, click Transform Data under the Home tab in Power BI Desktop. This opens the editor where you can load and transform your data. After opening, you can proceed with the following:

- **Add Data:** Load data into the Query Editor.
- **Perform Operations:** Apply transformations as required.
- **Prepare for Analysis:** Clean and shape data before using it in reports.



Importing Data

- In the Power Query Editor, click New Source to add data from sources like Excel, SQL or the web. After importing, you can apply necessary transformations to the data.
- For this example, we use an Excel file named “My movie list.xlsx”, and its data is shown below.



My Movie List



File Edit View Insert Format Data Tools Extensions Help



100% ▾



%

.0 ↵

.00 ➤

123

Arial ▾



D17 ▾



$\sum$ 8.6

	A	B	C	D
1	RANK	TITLE	GENRE	RATING
2	1	The Shawshank Redemption	Drama	9.3
3	2	The Godfather	Crime, Drama	9.2
4	3	Schindler's List	Biography, Drama, History	
5	4	Se7en	Crime, Drama, Mystery	8.9
6	5	It's a Wonderful Life	Comedy, Drama, Family	8.6
7	6	Goodfellas	Biography, Crime, Drama	8.7
8	7	Romeo + Juliet	Drama, Romance	6.7
9	8	12 Angry Men		9
10	9	Alien	Horror, Sci-Fi	8.4
11	10	Toy Story	Animation, Adventure, Comedy	8.3
12	11		Drama	8.7
13	12	To Kill a Mockingbird	Crime, Drama	
14	13	Rumble in the Bronx	Action, Comedy, Crime	6.8
15	14		Crime, Drama	7.5
16	15	Psycho	Horror, Mystery, Thriller	8.5
17	16	The Silence of the Lambs	Crime, Drama, Horror	8.6

On loading this data on the query editor it appears as:

 t ₃ RANK	A ₆ C TITLE	A ₆ C GENRE	1.2 RATING
1	1 The Shawshank Redemption	Drama	9.3
2	2 The Godfather	Crime, Drama	9.2
3	3 Schindler's List	Biography, Drama, History	null
4	4 Se7en	Crime, Drama, Mystery	8.9
5	5 It's a Wonderful Life	Comedy, Drama, Family	8.6
6	6 Goodfellas	Biography, Crime, Drama	8.7
7	7 Romeo + Juliet	Drama, Romance	6.7
8	8 12 Angry Men	null	9
9	9 Alien	Horror, Sci-Fi	8.4
10	10 Toy Story	Animation, Adventure, Comedy	8.3
11	11 null	Drama	8.7
12	12 To Kill a Mockingbird	Crime, Drama	null
13	13 Rumble in the Bronx	Action, Comedy, Crime	6.8
14	14 null	Crime, Drama	7.5
15	15 Psycho	Horror, Mystery, Thriller	8.5
16	16 The Silence of the Lambs	Crime, Drama, Horror	8.6
17	null	null	null

Common Data Transformations

1. Renaming Elements

The power query editor provides us with a variety of possible renaming. We can rename the data sources or tables, columns and queries. We will look into each of them one by one.

Rename Data Source

- You can rename a data source by right-clicking on it and selecting the Rename option.
- In this example, the source is renamed from Sheet1 to Movie Data. After renaming, the updated data is shown below.

Queries [1]



fx

= Table.Transform



Sheet1

- Copy
- Paste
- Delete
- Rename**
- ☒ Enable load
- ☒ Include in report refresh
- Duplicate
- Reference
- Move To Group
- Move Up
- Move Down
- Create Function...
- Convert To Parameter
- Advanced Editor
- Properties...



AB^B C TIT

1 The Sha

2 The Go

3 Schindl

4 Se7en

5 It's a W

6 Goodfe

7 Romeo

8 12 Ang

9 Alien

10 Toy Sto

11

12 To Kill a

13 Rumble

14

15 Psycho

16 The Sile

null

OUTPUT :

Queries [1]



Movie Data

Renamed source to "Movie Data"

AB C	TITLE	AB C	GENRE	1.2 RA
1	The Shawshank Redem			
2	The Godfather			
3	Schindler's List			
4	Se7en			
5	It's a Wonderful Life			
6	Goodfellas			
7	Romeo + Juliet			
8	12 Angry Men			
9	Alien			
10	Toy Story			
11				
12	To Kill a Mockingbird			
13	Rumble in the Bronx			
14				
15	Psycho			
16	The Silence of the Lam			
	null			

	Copy	
	Remove	
	Remove Other Columns	
	Duplicate Column	
	Add Column From Examples...	
	Remove Duplicates	
	Remove Errors	
	Change Type	▶
	Transform	▶
	Replace Values...	
	Replace Errors...	
	Split Column	▶
	Group By...	
	Fill	▶
	Unpivot Columns	
	Unpivot Other Columns	
	Unpivot Only Selected Columns	
	Rename...	

AB C	MOVIE NAME	AB C	GI
1	The Shawshank Redemption		Drama
2	The Godfather		Crime,
3	Schindler's List		Biogra

OUTPUT :

Renamed column to "MOVIE NAME"

Rename Query

- In the Query Settings pane, the Applied Steps section stores all the changes made as queries. You can rename these queries using the Query Editor.
- Here, the query “Renamed Columns” is renamed to “Columns Name Updation”. After renaming, it appears as shown.

The image shows two side-by-side screenshots from a data tool interface.

Left Screenshot (Query Settings):

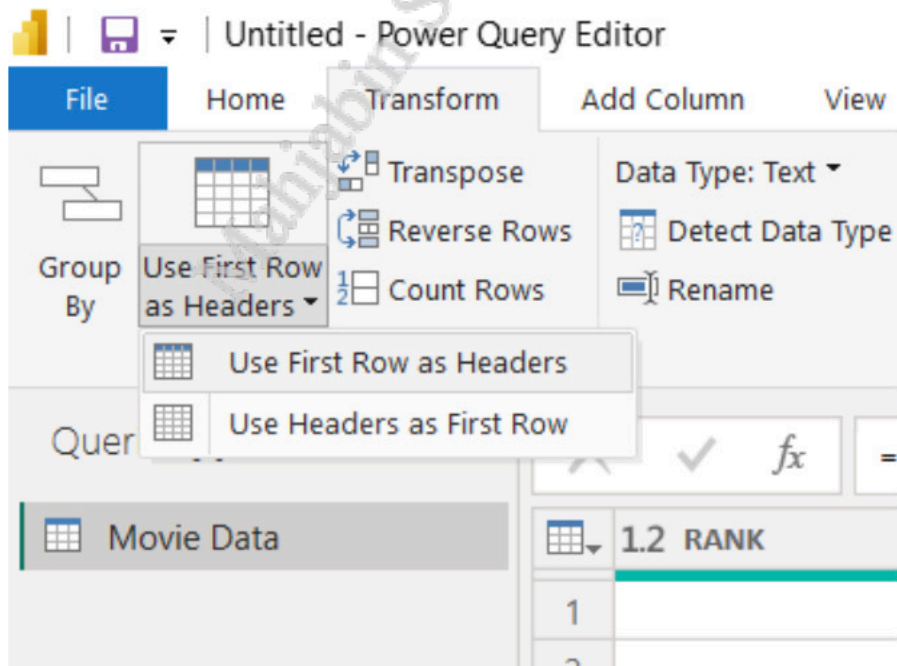
- Query Settings** (with a close icon)
- PROPERTIES** section:
 - Name:** A text box containing "Movie Data".
 - [All Properties](#)
- APPLIED STEPS** section:
 - A list of steps: Source, Navigation, Promoted Headers, Changed Type, and **Renamed Columns** (highlighted with a yellow background).
 - A context menu is open for the "Renamed Columns" step, showing options: [Edit Settings](#), **Rename** (highlighted with a yellow background), [Delete](#), and [Delete Until End](#).

Right Screenshot (OUTPUT):

- OUTPUT :**
- A list of items: Promoted Headers, Changed Type, and **Columns Name Updation** (highlighted with a grey background).

2. Make First Row As the Header in Power BI

- This operation is used to set the first row as column headers. It is available under the Transform tab in the ribbon.
- You can also use the option “Use Headers as First Row” to reverse this action if needed.



3. Change Data Types of a Column in Power BI

- You can change a column's data type by right-clicking the column, selecting Change Type, and choosing the required type (e.g., Whole Number to Decimal Number). You can also do this from the Home tab under Data Type.
- For example, the "RANK" column was changed from whole numbers (1, 2, 3) to decimal values (1.2, 2.3, 3.4).

	1 ² ₃ RANK	AB ₁ MOVIE NAME	AB ₂ GENRE
1			Drama
2			Crime, Drama
3			Biography, Drama, History
4			Crime, Drama, Mystery
5			Comedy, Drama, Family
6			Biography, Crime, Drama
7			Drama, Romance
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			

Copy
 Remove
Remove Other Columns
Duplicate Column
 Add Column From Examples...
Remove Duplicates
Remove Errors
Change Type
Transform
 Replace Values...
 Replace Errors...
 Group By...
Fill
 Unpivot Columns
Unpivot Other Columns
Unpivot Only Selected Columns
 Rename...
Move
Drill Down
Add as New Query

Decimal Number
Fixed decimal number
☒ Whole Number
Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...

OUTPUT :

1.2 RANK

Changed data type to "Decimal Number"

4. Format Operation in Power BI

The Format feature in the Transform tab is used to modify and clean text data in Power BI. It includes:

- **Change Case** : Convert text to uppercase or lowercase
- **Add Prefix/Suffix** : Add text at the beginning or end
- **Trim** : Remove extra spaces
- **Clean** : Remove non-printable characters

Query Editor interface showing the 'Format' dropdown menu. The menu options are:

- lowercase
- UPPERCASE
- Capitalize Each Word
- Trim
- Clean
- Add Prefix
- Add Suffix

The background shows a table with columns 'Rank' and 'Movie Title'.

Rank	Movie Title
1	
2	
3	
4	Se7en
5	It's a Wonderful Life
6	Goodfellas

UPPERCASE
THE SHAWSHANK REDEMPTION
THE GODFATHER
SCHINDLER'S LIST
SE7EN
IT'S A WONDERFUL LIFE
GOODFELLAS
ROMEO + JULIET
12 ANGRY MEN
ALIEN
TOY STORY
TO KILL A MOCKINGBIRD
RUMBLE IN THE BRONX

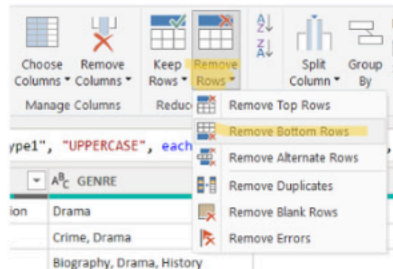
Formatting the text to UPPERCASE

5. Removal Operation in Power BI

Remove Row Operation

The Reduce Rows feature in the Home tab is used to remove unwanted rows in Power BI. It provides options to remove or keep specific rows based on your requirement. It includes:

- **Remove Rows** : Delete top, bottom, alternate, duplicate, blank or error rows
- **Keep Rows** : Keep only selected rows and remove the rest
- **Easy Operations** : All options work in a similar way (e.g., removing bottom rows)



Number of rows to be removed is 1

Row to be removed is number 17 :

16	26	The Silence of the Lambs	Crime, Drama, Horror	8.6	THE SILENCE OF THE LAMBS
17		null	null	null	null

Output : Row number 17 has been removed as follows

16	26	The Silence of the Lambs	Crime, Drama, Horror	8.6	THE SILENCE OF THE LAMBS
----	----	--------------------------	----------------------	-----	--------------------------

Removing the bottom row

Remove Column Operation

The Remove Columns feature is used to delete one or more columns in Power BI. It is available under the Home tab in the Manage Columns section. It includes:

- **Remove Columns** : Deletes the selected column
- **Remove Other Columns** : Keeps selected column and removes the rest
- **Example** : Removing the “RATING” column removes it from the dataset

Home Transform Add Column View Tools Help

New Query Data Sources Parameters Query Manage Columns

Remove Columns

Remove Other Columns

Table.ReplaceValue("#Removed Bottom Rows", null, "GENEALOGY", Replacer.ReplaceValue)

1	2	3 RANK	A ^B C TITLE	A ^B C GENRE	1.2	RATING
1	1		The Shawshank Redemption	Drama		9.3

Removing Column Rating

Output:

	1 ² 3 RANK	A ^B C TITLE	A ^B C GENRE
1		1 The Shawshank Redemption	Drama
2		2 The Godfather	Crime, Drama
3		3 Schindler's List	Biography, Drama, History

"RATING" column has been successfully removed.

6. Merge Operation in Power BI

The Merge Columns feature is used to combine multiple columns into one. It is available under the Add Column tab. It includes:

- **Select Columns** : Hold Ctrl and select multiple columns
- **Merge Columns** : Choose the merge option and set a column name
- **Add Separator** : Define a separator (e.g., “ ;) ”) between values
- **Example** : Columns “GENRE” and “RATING” are merged into a new column named “Merged”

Add Column | View | Tools | Help

Conditional Column | Index Column | Duplicate Column | Merge Columns | Extract | Parse | From Text | Statistics | Standard | Scientific | From Number | Trigonometry | Rounding | Information | From Date & Time | Date | Time | Duration | Text Analytics

X ✓ fx = Table.AddColumn(#"Removed Bottom Rows", "Merged", each Text.Combine({[GENRE], Text.From

1.2	RANK	A ^B _C	MOVIE NAME	A ^B _C	GENRE	1.2	RATING	A ^B _C
1		1	The Shawshank Redemption		Drama		9.3	TH
2		2	The Godfather		Crime, Drama		9.2	TH

Merge Columns

Choose how to merge the selected columns.

Separator

--Custom--

}

New column name (optional)

Merged

OK

Cancel

Merging Columns GENRE and RATING

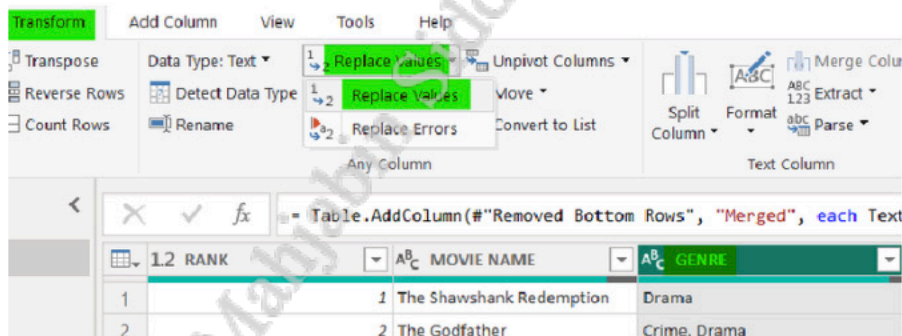
Output:

A ^B _C	Merged
	Drama ;) 9.3
	Crime, Drama ;) 9.2
	Biography, Drama, History
	Crime, Drama, Mystery ;) 8.9
	Comedy, Drama, Family ;) 8.6
	Biography, Crime, Drama ;) 8.7
	Drama, Romance ;) 6.7
	9
	Horror, Sci-Fi ;) 8.4
	Animation, Adventure, Comedy ;) 8.3
	Drama ;) 8.7
	Crime, Drama
	Action, Comedy, Crime ;) 6.8
	Crime, Drama ;) 7.5
	Horror, Mystery, Thriller ;) 8.5
	Crime, Drama, Horror ;) 8.6

Merged Column

7. Replace Values Operation in Power BI

Replace values operation replaces some specific value to our desired value. It's present in the "Transform" bar as "Replace Values". Here, we have replaced "null" to "geeksforgeeks" for column "GENRE".



The screenshot shows the Power BI Transform ribbon with the 'Replace Values' button highlighted. The 'Value To Find' is 'null' and the 'Replace With' is 'geeksforgeeks' for the 'GENRE' column. The data table below shows the results of the operation.

	1.2 RANK	AB MOVIE NAME	AB GENRE
1		1 The Shawshank Redemption	Drama
2		2 The Godfather	Crime, Drama

Replacing values of column "GENRE"

Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

geeksforgeeks

> Advanced options

Output:

A ^B _C GENRE
Drama
Crime, Drama
Biography, Drama, History
Crime, Drama, Mystery
Comedy, Drama, Family
Biography, Crime, Drama
Drama, Romance
geeksforgeeks
Horror, Sci-Fi
Animation, Adventure, Comedy
Drama
Crime, Drama
Action, Comedy, Crime
Crime, Drama
Horror, Mystery, Thriller
Crime, Drama, Horror

Successfully Replaced

8. Split Column Operation in Power BI

The Split Column feature is available under the Transform tab and is used to divide a column into multiple columns. It includes:

- **Split Options** : Split by delimiter, number of characters, or position
- **Example** : Column “Merged” is split using delimiter “ ;) ”
- **Output** : Creates new columns like “Merged.1” and “Merged.2”



Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

--Custom--

;

Split at

- ☐ Left-most delimiter
- ☐ Right-most delimiter
- ☒ Each occurrence of the delimiter

> Advanced options

Quote Character

"

☐ Split using special characters

Insert special character

Before

A ^B _C Merged	
Drama ;)	9.3
Crime, Drama ;)	9.2
Biography, Drama, History	
Crime, Drama, Mystery ;)	8.9
Comedy, Drama, Family ;)	8.6
Biography, Crime, Drama ;)	8.7
Drama, Romance ;)	6.7
9	
Horror, Sci-Fi ;)	8.4
Animation, Adventure, Comedy ;)	8.3
Drama ;)	8.7
Crime, Drama	
Action, Comedy, Crime ;)	6.8
Crime, Drama ;)	7.5
Horror, Mystery, Thriller ;)	8.5
Crime, Drama, Horror ;)	8.6

After

A ^B _C Merged.1	1.2 Merged.2
Drama	9.3
Crime, Drama	9.2
Biography, Drama, History	null
Crime, Drama, Mystery	8.9
Comedy, Drama, Family	8.6
Biography, Crime, Drama	8.7
Drama, Romance	6.7
9	null
Horror, Sci-Fi	8.4
Animation, Adventure, Comedy	8.3
Drama	8.7
Crime, Drama	null
Action, Comedy, Crime	6.8
Crime, Drama	7.5
Horror, Mystery, Thriller	8.5
Crime, Drama, Horror	8.6

After Split Scenario

9. Pivot Column Operation in Power BI

The Pivot Column feature is used to convert rows into columns. It is available under the Transform tab in Power BI. It includes:

- **Row to Column Conversion :**
Transforms row values into columns
- **Aggregation Options :** Default is Sum, but can be changed to Min, Max or Don't Aggregate
- **Flexible Usage :** Choose aggregation based on requirement

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Table

Transpose Reverse Rows Count Rows

Data Type: Decimal Number 1 2 Replace Values Unpivot Columns

Detect Data Type Fill Pivot Column Move Convert to List

Rename Any Column

Data for pivot operation:

	1.2 Field 1	1.2 Field 2
1	1.1	2
2	2.6	9
3	4.3	2
4	2.6	3
5	3.9	2

Pivot Column

Use the names in column "Field 2" to create new columns.

Values Column ⓘ

Field 1

Advanced options

Aggregate Value Function

Sum

[Learn more about Pivot Column](#)

Output :

OK

	1.2 2	1.2 9	1.2 3
1	9.3	2.6	2.6

Pivot column "Field 2"

10. Unpivot Column Operation in Power BI

The Unpivot Column feature is used to convert columns into rows, which is the opposite of pivot. It gathers similar values under a single column with corresponding labels. It includes:

- **Column to Row Conversion** : Converts multiple columns into rows
- **Organizes Data** : Combines values under one label for better structure
- **Opposite of Pivot** : Reverses the pivot operation

Transform Add Column View Tools Help

Transpose Reverse Rows Count Rows

Data Type: Decimal Number 1 2 Replace Values Fill Pivot Column Any Column

Unpivot Columns

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

fx = Table.TransformColumnTypes(Source,{{"Field 1", type

	1.2 Field 1	1.2 Field 2
1	1.1	2
2	2.6	9
3	4.3	2
4	2.6	3
5	3.9	2

Unpivot operation implemented on Field 1 and Field 2 both

Output:

	Attribute	Value
1	Field 1	1.1
2	Field 2	2
3	Field 1	2.6
4	Field 2	9
5	Field 1	4.3
6	Field 2	2
7	Field 1	2.6
8	Field 2	3
9	Field 1	3.9
10	Field 2	2

On unpivoting both the columns